



MIMIR-V: SCALING LLM SERVICES FOR U.S. ARMY

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PROJECT OVERVIEW AND GOALS

OUR TEAM

1LT Roberts, Jonathon
Project Manager

SGT Cowgill, Ryan
RAG Lead

CW3 Vettore, Nicholas
Systems Lead

1LT Lewis, Ka'Ron
Kubernetes Lead

CPT Fairbank, Nate
AI2C Sponsor/ Mentor

CPT Doyle, Aidan
MCDS Partner

Mimir-V Project Goals

Goal

Provide complex document interaction with LLM service to improve staff workflow at every echelon

Audience

Service up to 500 service members within a Division level unit

Deployment

Improve service for:
- Scalability
- Modularity
- Platform-Agnostic

Approach

Validate system changes with quantitative metrics at every step of the way to prove value-add

PROBLEM

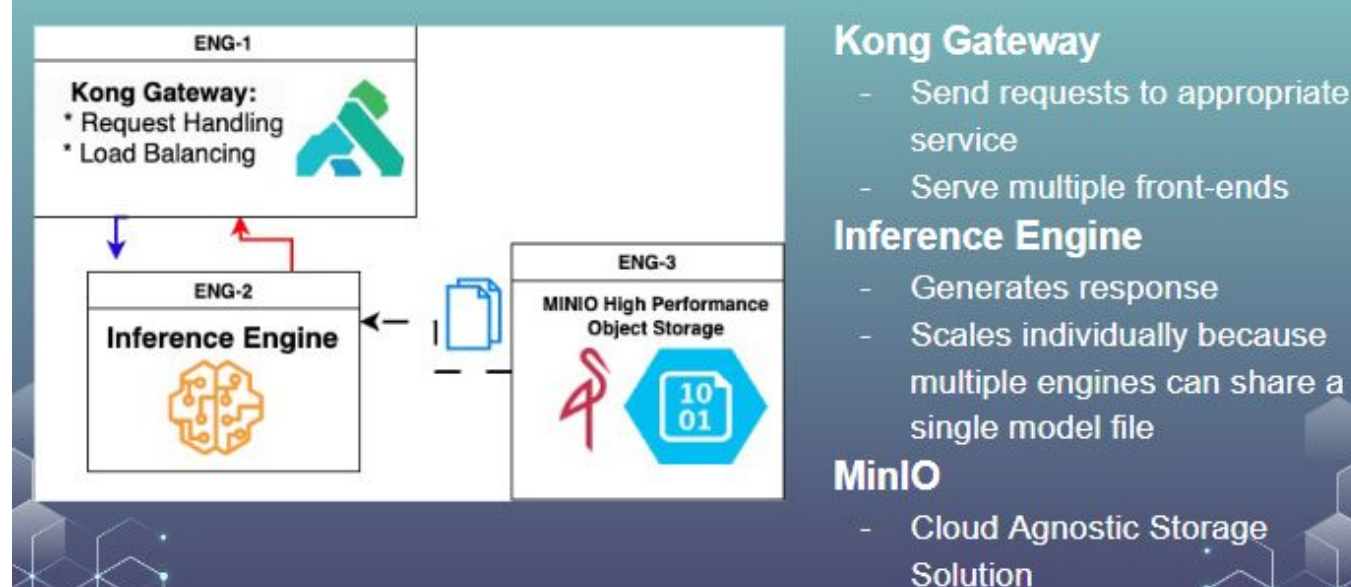
Commercial LLM services are inaccessible on military networks, unable to assist with Controlled or Classified information, and untraceable

CURRENT SOLUTION

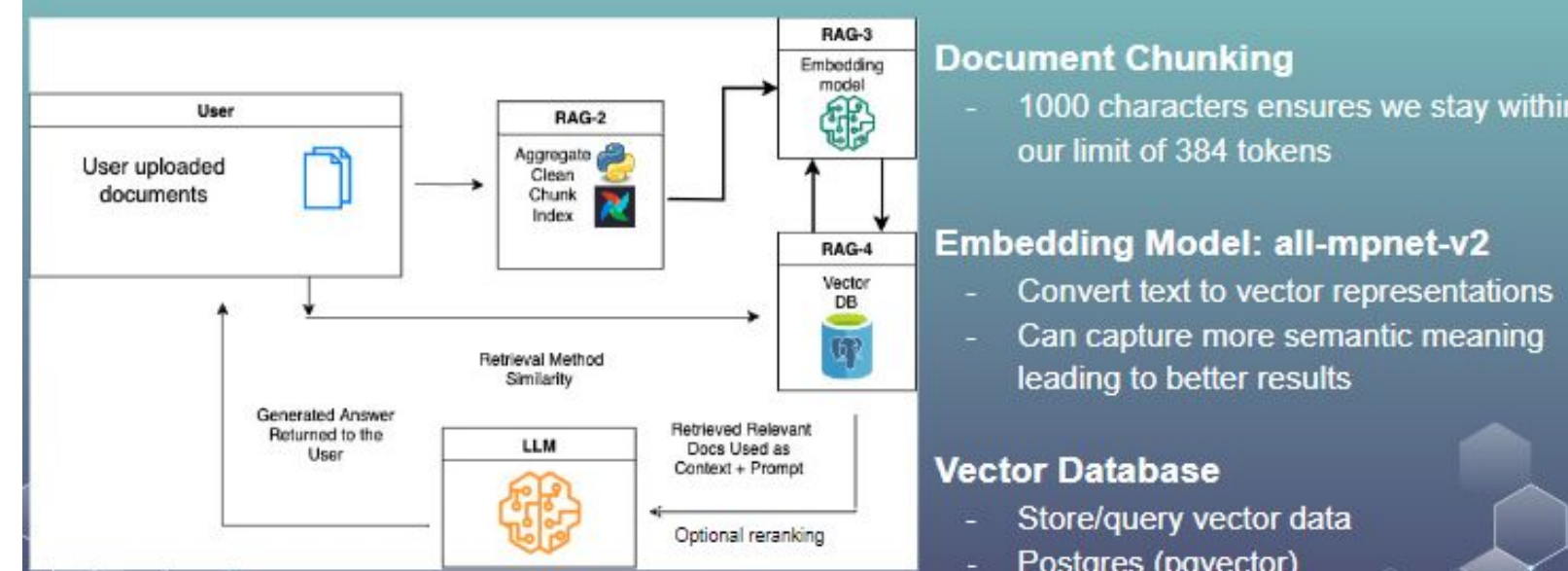
MIMIR-baseline solves the accessibility and classified information problem, but can only serve 10 users concurrently with no work document interaction

ADAPTABLE SYSTEM DESIGN

Scalable Inference Engine



Retrieval Augmented Generation (RAG) Design



Orchestration and Observability



EXPERIMENT-DRIVEN DEVELOPMENT

EXPERIMENT DESIGN

EXPERIMENT 1: Can the Army effectively deploy and scale an LLM service on the existing Army hardware stack?

Load Generation

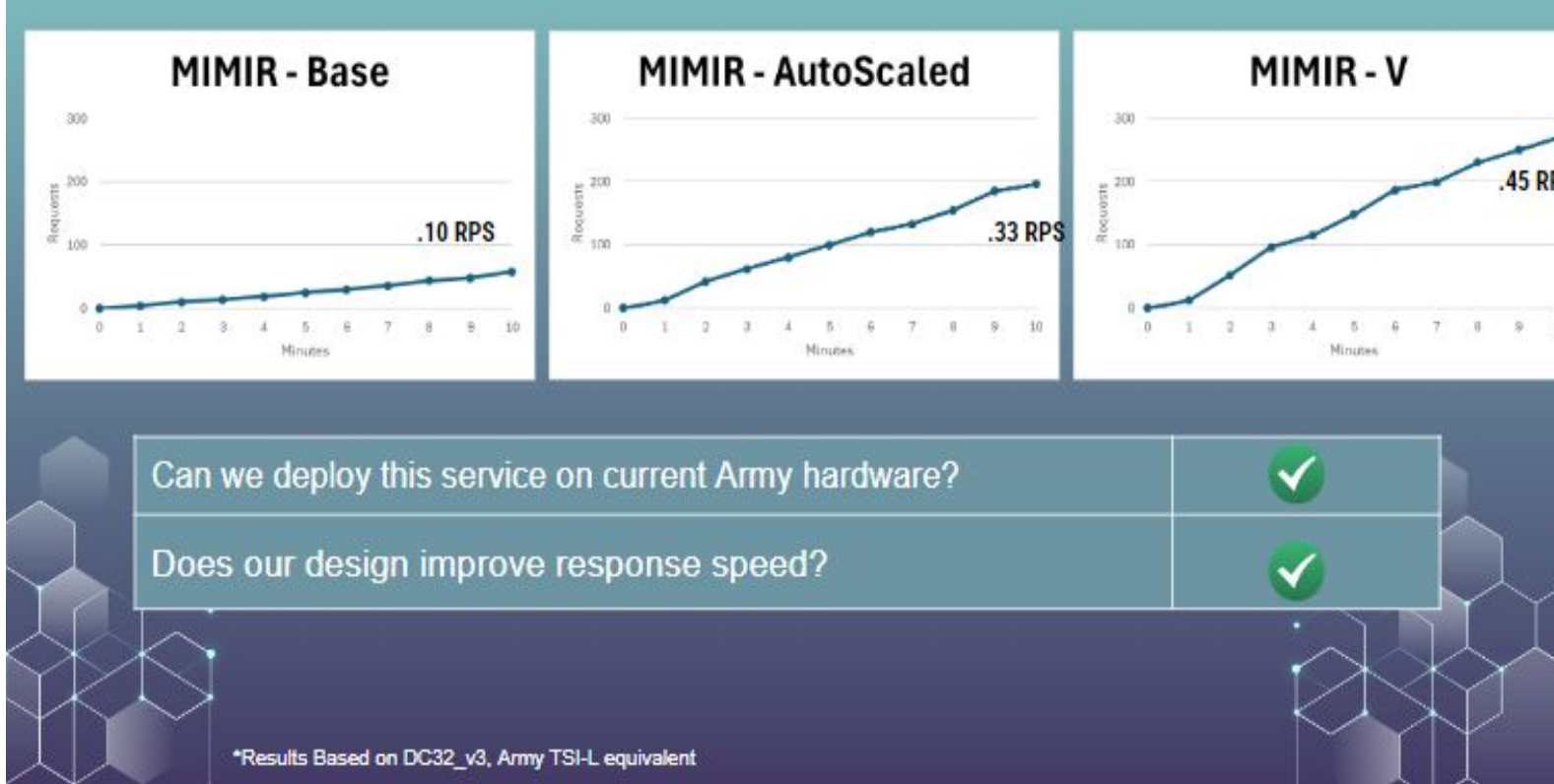
- Simulate load of 150 users with frequent requests
- Conducted on Azure equivalent of Army Hardware
 - B16ms, DC32_v3, D64d_v4

EXPERIMENT 2: Does Army-specific content affect contextual retrieval and response generation?

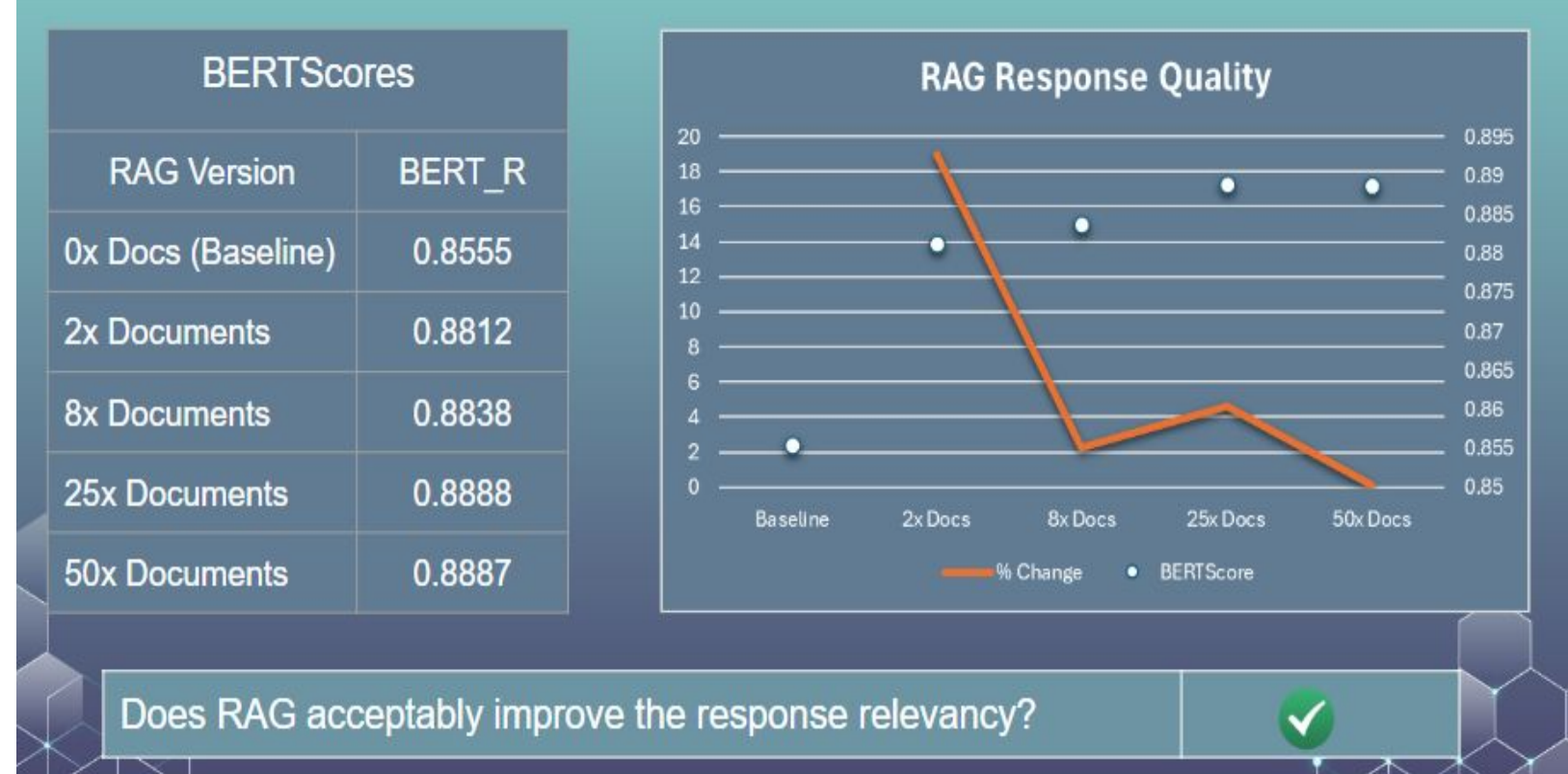
Dataset Design - 100 Total Questions

- 4x Army knowledge domains: Infantry, Intelligence, Tactical Signal, and SOF
- 25x questions per domain along with an associated "ground truth", doctrinally-referenced answer
 - Army Doctrine pulled directly from Army Pubs

Experiment Results 1

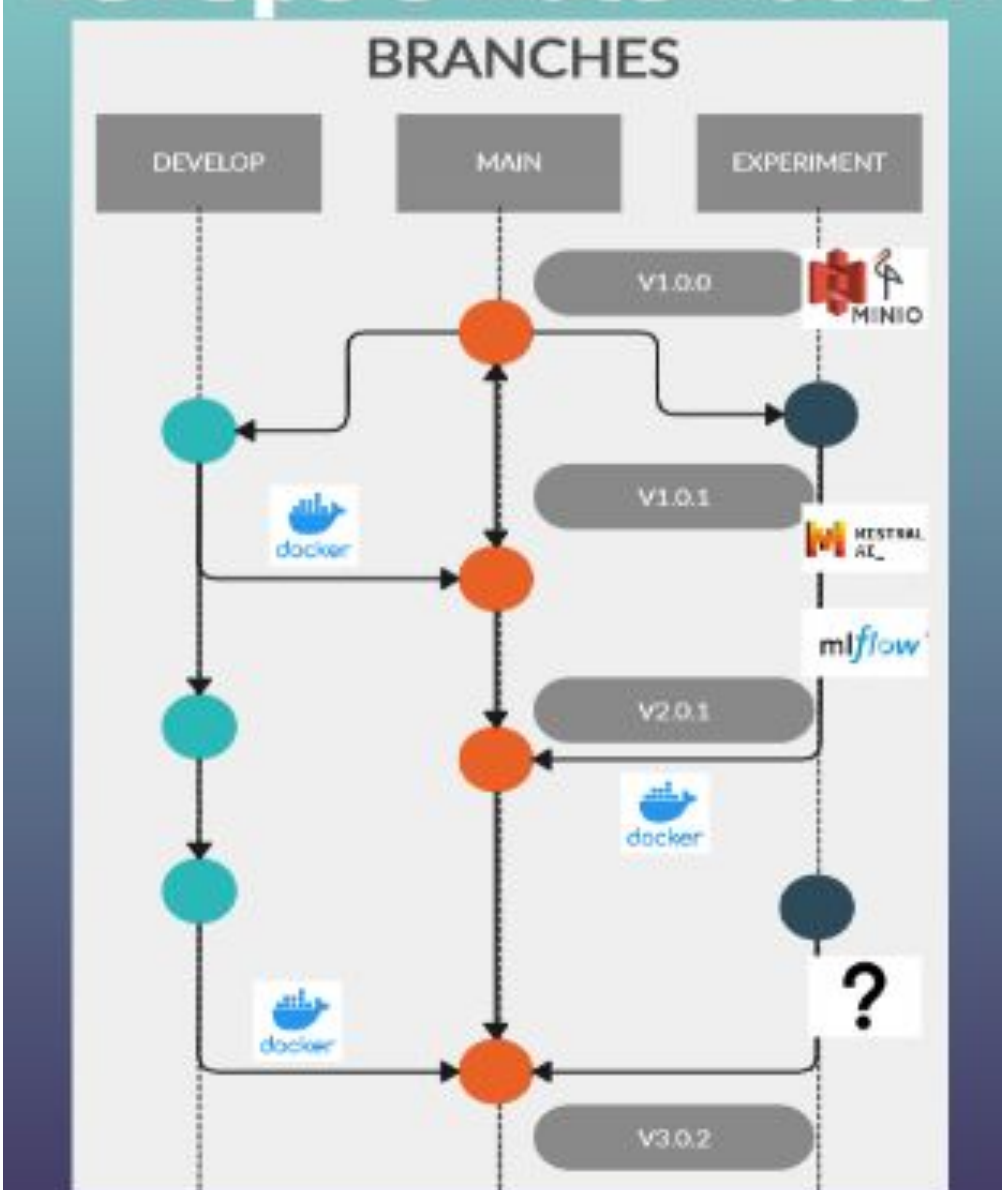


Experiment Results 2



AUTOMATED AGILE DEPLOYMENT

DevOps & Automation



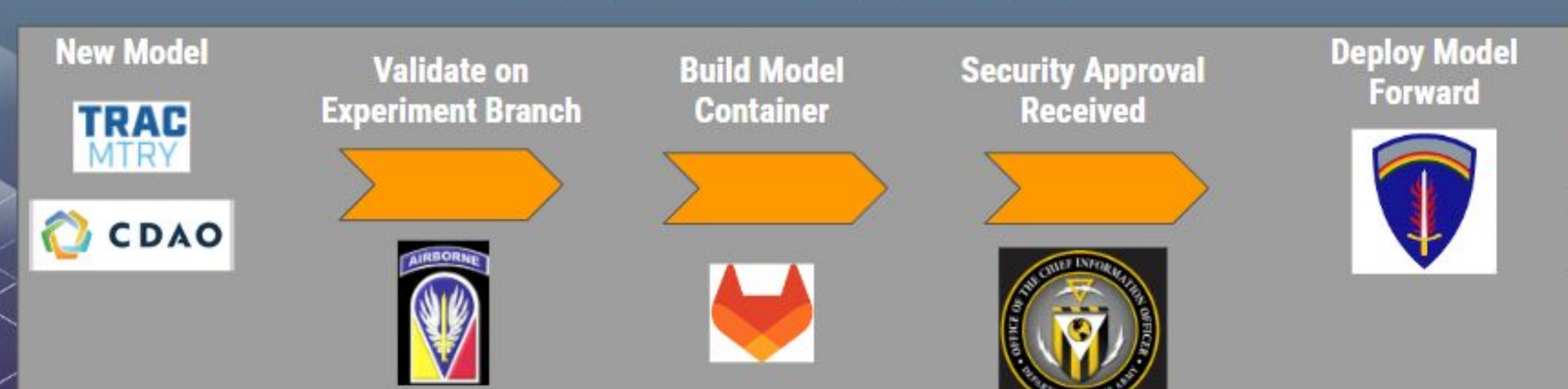
Built for Enterprise and Tactical Agility

AI2C Success Metric: # of users
Can Mimir-V continue to add users?

Attract Users with Agility

LLM technology will constantly improve, and our modular system can keep the DoD at the technological edge.

Example Tactical Use Case



AI Technicians 2023-2024 - Capstone Project
supervised by CPT Nathan Fairbank, CPT Aidan Doyle